Foreword

Global investment in climate finance has been on the decline since its peak in 2022 despite the increasing threat that climate instability poses to our lives, communities, and businesses.

Energy majors are shifting away from cleantech, a trend likely to be further exacerbated by shifting political landscapes.[1, 2]

Over my years supporting Project Frame, it has become evident that while many participants join out of a drive to secure a livable climate, the majority see greenhouse gas (GHG) impact analysis as key to their portfolio success. In our most recent survey, over 90 percent of respondents reported seeking above-market rate returns and indicated that aside from climate concerns, their primary motivator was firm competitiveness. It has also been my experience that limited partners and growth equity are becoming increasingly aware of the need for GHG impact analysis to mitigate further risk and exposure to their existing assets. Even with these trends, investment isn't transitioning fast enough to reach emissions reduction targets in time to avert significant harm within our lifetime. As climate disasters become more frequent and intense, forward-looking emissions assessment will be essential to remain relevant amidst evolving market demands in our new climate reality.

In response, Project Frame has grown to over 900 participants within three years, representing more than 360 firms with $670.1 billion in venture capital and private equity assets under management. The 2023 methodology has been viewed more than 7,500 times, including over 1,200 downloads. Powered by Prime Coalition's philanthropic support and volunteer participation, the rapid increase in enthusiasm highlights investors' demand for actionable approaches to assessing GHG impact.

"I am overwhelmed with gratitude for the community of investors and thought leaders dedicated to accountable and transparent climate investing that Project Frame has fostered. The need for impactful, transformative climate solutions is more pressing than ever and I believe deeply that Project Frame's guidance will help many make the most impactful investment decisions they're able to make."

SARAH KEARNEY
Executive Director
Prime Coalition

1 *https://www.ctvc.co/32bn-and-30-drop-as-market-hits-pause-in-2023/?ref=ctvc-by-sightline-climate-newsletter*
2 *https://www.ctvc.co/fuel-disclosure-from-energy-majors-earnings-221/?ref=ctvc-by-sightline-climate-newsletter*

This is further exemplified by Project Frame's recognition from Ceres, Global Financial Alliance for Net Zero, Global Impact Investing Network, World Business Council for Sustainable Development, and most recently, the Government of Japan. Looking forward, we are encouraged to see Project Frame's learnings embedded into emissions impact assessment tools like Climate Point, CRANE, Koi, and MoreScope, while we continue to iterate improvements within our newest working groups composed of more than 50 investors across 4 continents.

This latest iteration of the Project Frame methodology—along with a new library of accompanying case studies and supplemental materials— demonstrates that GHG impact forecasting and reporting is both an art and a science. Those who practice it are vanguards, competing to find and support necessary solutions for a carbon-constrained future. That is why we have designed the methodology to be practical, built and implemented by investors alongside climate experts who understand the importance of transparency and accountability for industry credibility.

We extend our deepest gratitude to our philanthropic supporters, steering committee, working groups, community members, and team Prime for their unwavering dedication to our mission and the development of this publication. Now more than ever, it is vital for philanthropy and the private sector to commit to driving the transformation needed for our businesses, communities, and planet to thrive.

Keri Browder
Director of Project Frame & Impact Accountability
Prime Coalition

"Mitigating climate change is non-negotiable; it is therefore critical that we leverage the most robust, material, scientific and precise measurement practices to meaningfully measure and articulate the impact of our/our portfolio's work. We deeply respect Project Frame's considered approach in this regard and are privileged to be supporting it and its members on this important mission."

MAYA CHANDRASEKARAN
Co-Founder & Managing Partner
Green Artha

"For climate investment, GHG impact quantification is absolutely key for our opportunity screening. This Frame methodology update provides further standards to help us refine and improve our approach."

MATTHEW HARWOOD
Chief Strategy Officer
Climate Investments

About Project Frame

Project Frame (Frame) is a nonprofit program, convened by Prime Coalition, built to organize investors around forward-looking emissions impact methodology and reporting best practices. Our mission is to mitigate climate change by demystifying climate investing and improving Impact Measurement & Management (IMM) to drive capital towards the best possible climate solutions while galvanizing a network around transparency, accountability, and collaboration.

Our goals include:

1. Helping investors efficiently identify investments with the highest potential to achieve GHG mitigation.
2. Demystifying the process to encourage consistent procedures for including forward-looking GHG impact assessment in investment decision-making, centering transparency in order to discourage bad actors and greenwashing.
3. Facilitating collaboration to develop common terminology and best practices to encourage a shared understanding that aligns efforts across different stakeholders, fostering a more coordinated approach to addressing climate change.

About Prime Coalition

Prime Coalition is a non-profit organization on a mission to unlock catalytic capital and change the future of climate finance.

Since 2015, Prime has mobilized over $317MM in catalytic capital with 430 partners to back 39 early-stage ventures. Through its steering capital strategy, including Prime Impact Fund, Azolla Ventures, and Trellis Climate, Prime builds and implements impact-first investing teams toward acute gaps in climate finance that commercial capital cannot fill.

With its influencing capital strategy, Prime offers a range of field-building resources, including Project Frame, the open-source CRANE tool, and the Catalytic Capital Intermediation Resources Library.

"Investing wisely to accelerate the energy transition is more important than ever. The Project Frame methodology guides investors in evaluating the climate impact of climate solutions companies, which makes the impact investing process more efficient and effective."

DAN MILLER
Managing Director
The Roda Group

"This report makes a significant contribution towards improving impact assessment throughout our sector and is another example of Project Frame's contribution towards better impact practices."

ADAM JAMES
Partner, Head of Customer Experience & Impact
Energy Impact Partners

SCOPE OF ANALYSIS

PURPOSE

Project Frame's methodology is designed to help investors assess the GHG impact of climate solutions. It is meant to align investors with solutions from pre-seed stage into growth equity that are on a path to sustainable, scalable impact. It also seeks to establish common language to help investors, entrepreneurs, and NGOs communicate transparently about the GHG impact of their solutions.

The current version of the methodology is written for investors who have moved beyond early learning in GHG impact assessment, building upon Frame's original 2023 methodology, "Pre-Investment Considerations: Diving Deeper Into Assessing Future Greenhouse Gas Impact."

What Is and Is Not Analyzed	
Greenhouse Gases (GHGs)	The Frame methodology focuses on GHG metrics alone and does not currently provide recommendations on other environmental or social metrics. However, it can accommodate other dimensions guided by the investor.
Comparing a Solution to its Incumbent(s)	The methodology is designed to compare GHGs of a solution and the appropriate incumbent or status quo. For example, the solution unit is the smallest replicable instance of a solution sold, such as a heat pump, while the incumbent unit is the equivalent product it would displace. This is the basis of our definition of "impact" and distinct from GHG footprinting, which typically assesses absolute amounts of past GHGs emitted by a product, solution, or company. The GHG impact formula can also meet the needs of investors assessing climate solutions that do not have an incumbent, such as in carbon removal. For more, read "Classification System for Climate Solutions."
Scalability	The methodology is meant to help investors decide whether to invest in a proposed climate solution that achieves GHG impact by scaling, and thus taking market share away from the incumbent. However, scalable technologies can also cause rebound effects that have no or negative impact. We discuss this in "Rebound Effects."
Assessing the Past & Future	The current Frame methodology covers both forward and backward looking GHG impact assessment. As solutions evolve, methodologies must evolve. Frame's methodology is meant to help investors achieve the most accurate analysis possible based on the available data for the solution stage and use historical data when it is available to inform investment decision-making and reporting.

Critical Workflows	
Assessing for Materiality	An analyst may encounter a material source of difference at any life cycle stage. Frame emphasizes that a cradle-to-grave system boundary is essential in qualitative analysis. But investors in early-stage solutions must parse through dozens, if not hundreds, of solutions to deploy capital in a short period of time. How can they efficiently review all stages of the life cycle, knowing that it is unrealistic to quantify every single potential source of difference between a solution and incumbent and that some sources of difference massively outweigh others? What sources of unit emissions are material, or "worth quantifying?" This question forms the basis of a step in net unit impact analysis, meant to help analysts think through which sources of unit emissions are material before spending time quantifying sources that are unlikely to prevent the solution from clearing a firm's GHG impact targets. Depending on investor feedback, we may apply the concept of materiality to other steps in the assessment process. This use of materiality is different from others often heard in the investing context, such as whether environmental or social factors are material to a company's financial performance.
Downselection	There are plenty of proposed climate solutions. Among them, fewer have a significant unit impact and far fewer of those have the realistic potential to scale. Even within that smallest pool, there are not enough investment dollars available to support all. How do investors efficiently filter—or downselect—solutions to drive capital towards solutions that have the highest probability of clearing both impact and business targets? The Frame methodology starts with market segmentation and overall business analysis to ground impact analysis in business realities, meshing with traditional downselection processes.

CLASSIFICATION SYSTEM FOR CLIMATE SOLUTIONS

Frame describes a climate solution as an intervention, product, or service that intends to and shows evidence it can achieve GHG impact. Project Frame defines impact as a real-world change caused or enabled by an organization (based on the goods or services produced). Impact can be positive or negative; intended or unintended; direct or indirect; or incremental or systemic.

GHG impact is a change in GHG emissions caused by an organization. Frame defaults to GHG impact implying a “positive” outcome unless otherwise stated. In addition, since Frame’s focus is early-stage, an organization may represent a single solution. Frame currently classifies climate solutions by impact and pathway types.

Impact Types: How the Solution Broadly Achieves GHG Impact	
Avoiding Emissions	Avoids future emission by replacing or taking market share over time from an incumbent responsible for higher emissions. Most climate solutions today fit within this category.
Removing Emissions	Removes carbon from the atmosphere. In this case, there might not be an incumbent to compare the solution to, but that doesn’t automatically imply that its impact will be positive. All new sources of emissions must be considered, as well as potential ways in which such technologies could trigger consequences elsewhere in the system. See “Action and Reaction Across Systems” for more.

Pathway Types: How a Solution Functions Within a System of Integrated Solutions	
Direct	In direct solutions, the positive GHG impact would not occur without them. • **Product:** Complete solutions that can be purchased and used to directly achieve GHG impact. Examples include an electric vehicle (EV), heat pump, or sustainably produced food. • **Component:** Critical parts of an overall solution that contribute significantly to its GHG impact. The overall impact depends on how the product containing the component is used. Examples include an EV battery, more efficient motor, or recycled materials
Facilitating	Facilitating solutions enable or improve the effectiveness of direct solutions. They do not directly achieve GHG impact on their own, but enhance direct solutions. Facilitating solutions may: • **Affect the unit impact of a direct solution,** such as software that significantly improves the efficiency of renewable energy technologies or reduces waste or energy use in manufacturing processes. • **Affect volumes of a direct solution**, such as a green marketplace that accelerates the deployment of electric vehicles by connecting consumers with suppliers or financing.

Why Does Classification Matter?

To achieve GHG impact, no solution works entirely on its own. Even direct product solutions— the solution pathway upon which the Frame methodology is built—achieve GHG impact through components and an ecosystem of suppliers, marketplaces, policies, and more. Frame's classification system is meant to help investors systematically structure quantification for every combination of solution impact and pathway type, typically by focusing first on the most proximate source of impact: the direct product. From there, investors modulate unit impact, volumes, or adjustment factors according to the solution they may be considering for investment.

ADDITIONAL RESOURCES

Scope of Analysis Worksheet
A scope of analysis clearly defines what is and is not the focus of analysis. This worksheet helps investors clarify their own scopes as they communicate with investors and LPs regarding their methodologies.

Investor Profiles
Through Investor Profiles, investors share their impact assessment approaches in a consistent way. While investors conform to Frame's definition for GHG impact, we welcome them to share approaches to other forms of assessment that are not within Frame's current scope of analysis, such as on other environmental, social, and governance topics.

Case Studies
Frame's growing library of emissions impact assessment case studies showcase our methodology in action.

Assessing Environmental Impact: A Framework Comparison
Wondering how GHG impact is different from GHG footprinting? This article gives a comparison at a glance.

Potential GHG Impact

Potential GHG impact estimates the impact a proposed climate solution could have based on a standardized growth trajectory that assumes the proposed solution takes over the Serviceable Obtainable Market (SOM).

This analysis is valuable for early-stage solutions that could alter emissions patterns and is most applicable when looking at early-stage solutions with limited information. Potential impact analyses are longer-term and often align with global scenarios by [the International Energy Agency (IEA)](). Potential impact analyses carry a wide band of uncertainty.

Planned GHG Impact

Planned GHG impact estimates the future GHG impact that can be realistically achieved, considering the specific solution's unit impact alongside its detailed forecasts of sales and/or the installed base. It is more applicable to companies with realistic business plans that reflect the solution's specific market segments and incorporate fleet effects for solutions with long operational lifespans. Planned impact analyses typically align with timeframes established by business plans.

Realized GHG Impact

Realized GHG impact is intended to serve as the most accurate representation of GHG impact within a designated historical timeframe. At this stage, solution and incumbent unit emissions should be built off of GHG footprint, validated with real-world data. It is calculated by multiplying the most up-to-date measure of unit impact by historical volumes, such as sales and/or the installed base, assuming sales have occurred. This analysis provides a critical feedback loop by comparing actual data with planned forecasts, enabling both the company and investment analyst to refine their strategies and improve future projections.

Net Unit Impact (CO_2e/Unit) (with SOM parameters) **x** **SOM** for technology class Units/Year **=** **Potential Impact** Potential GHG Impact estimates the impact a proposed climate solution could have based on a standardized growth trajectory that assumes the proposed solution takes over the Serviceable Obtainable Market (SOM).

Net Unit Impact (CO_2e/Unit) (with commercial forecast parameters) **x** **Commercial Forecast** for specific company Units/Year **=** **Planned Impact** Planned GHG Impact estimates the future GHG impact that can be realistically achieved, considering the specific solution's unit impact alongside its detailed forecasts of sales and/or the installed base.

Net Unit Impact (CO_2e/Unit) (with historical volume parameters) **x** **Historical Volumes** for specific company Units/Year **=** **Realized Impact** Realized GHG impact is intended to serve as the most accurate representation of GHG impact within a designated historical timeframe. At this stage, solution and incumbent unit emissions should be built off of GHG footprint, validated with real-world data.

METHODOLOGY CHAPTERS, AT A GLANCE

Market Segmentation

Volumes and unit impact are not isolated variables. For example, where the solution is sold or how it is used affects its unit emissions. In this chapter, we discuss how investors integrate GHG analysis into due diligence, including market sizing and segmentation. This helps to drive better investment decisions. This also reflects the downselection process, which typically begins with assessing scalability and market context.

Net Unit Impact

Net unit impact, or "unit impact," measures the difference in emissions between a solution and an incumbent across their life cycles and in alignment with the market segments. These differences, called Solution Effects, are summed to a net unit impact value. Assumptions in net unit impact calculations change over time. We encourage analysts to think qualitatively through the materiality of Solution Effects before quantifying them.

Volumes

In this chapter, we go deeper into volumes, which represent an annual number of solution units over a specified time frame, such as number of sales or units in operation. Sizes of markets and market segments can be quantified from top down or bottom up.

GHG Impact

In this chapter, we provide further detail on the three approaches to assessment—potential, planned, and realized impact—and explain how volumes combine with net unit impact to visually represent the anticipated GHG impact of a solution over time.

New Considerations

The GHG impact equation covers all variables that directly affect solution unit emissions, incumbent unit emissions, and solution volumes. But investors have raised the following additional considerations for further incorporation and future guidance:

- **Optional Adjustment Factors:** Adjustment factors are variables of GHG impact that are separate from and shouldn't be blended into any underlying variables associated with the unit impact or volumes of the direct product solution being analyzed. They are considered optional at this time. In each section below, we also discuss challenges with the current assessment methods.
    - *Value Chain Attribution:* With this adjustment, investors attribute portions of GHG impact to multiple direct component solutions along the value chain.
    - *Capitalization Attribution:* With this adjustment, investors apportion GHG impact among owners of the solution they have invested in.
- **Rebound Effects:** Rebound effect refers to the ways in which technology that increases efficiency through the use of a resource, such as energy, can cause the resource to be used more. In theory, this can lessen or cancel out the positive environmental benefits of efficiency.
- **Action & Reaction Across Systems:** In this category, we consider how and/or why investors may need to take into account the impact of their solution on another part of the system, such as a different solution or service.

Geography

Assessments may differ by region, due to varying local market dynamics, regulatory environments, natural conditions, and customer preferences. Examples of key considerations include:

- **Grid intensity:** The feasibility and impact of solutions can vary significantly depending on grid intensity. For example, an electric vehicle may have a higher impact in a region with a cleaner grid compared to one with a coal-dominated energy mix.
- **Resource availability:** Resources vary by region, such as the presence of materials that affect the way in which you make the solution and how energy can be produced. For example, solar solutions are more effective in sunnier regions, while wind solutions might be better suited for windy areas.
- **Temperature dynamics:** The GHG impact of products like furnaces and air conditioners will differ based on climate. Cold climates demand more heating, while warm climates require more cooling, affecting overall emissions.

Different Incumbents

Different markets may have different incumbents. For example, a heat pump may replace both natural gas furnaces and electric baseboard heaters. Similarly, an electric vehicle (EV) might replace both gasoline-powered cars and hybrid vehicles, each with distinct emissions profiles.

Usage Profiles

Many climate solutions have distinct usage profiles, or ways of being used, depending on duration, intensity, or frequency of use. For example, an electric truck might be used both as a personal vehicle and within commercial fleets. Personal use might involve shorter, less frequent trips, while commercial fleet use could mean longer, more frequent journeys with higher loads.

Product Variations

Products can vary by size, how they are packaged to be sold—such as standard and deluxe product—and other important configurations. For instance, a standard model might appeal to cost-conscious consumers, while a deluxe version might target premium segments with higher performance expectations.

5 Steps of Analysis

Solution Effects analysis is structured into five steps intended to produce a narrowed down set of material effects that advance to quantification of net unit impact.

LIST SOLUTION EFFECTS	ANALYZE ASSOCIATED INCUMBENT	ANALYZE ASSOCIATED SOLUTION	COMPARE SOLUTION & INCUMBER	ASSESS FOR MATERIALITY

Step 1

List Solution Effects

- **Solution Effect (narrative):** Name or summarize each source of difference in emissions between the solution and incumbent, and explain why they are different.
- **Life Cycle Category (limited choice):** Identify the stage of the life cycle relevant to the effect (e.g., production, use, disposal).
- **Emissions Source (narrative):** Detail the specific source of emissions (e.g., fuel combustion for ICE vehicles, charging emissions for EVs).
- **Sector (limited or multiple choice):** Provide context by identifying the primary industry or sector involved (e.g., transportation, energy).
- **Geography (limited choice):** Define where the emissions occur and note differences between the incumbent and the new solution.
- **Emissions Type (limited choice):** Identify which GHGs are affected (e.g., CO_2, CH_4, NH_3) and note any other relevant environmental impacts. This is essential for clarifying whether and how to incorporate different global warming potential (GWP) values.
- **Frequency (limited choice):** Classify the emissions effect as one-time, intermittent, or recurring to understand the regularity and predictability of the emissions.

Step 2

Analyze the Associated Incumbent

Isolate and further evaluate emissions of the incumbent as it relates to the solution effect, including estimated amounts of emissions associated with the effect, trends, and rationales for the incumbent.

- **Estimated Emissions Amount (value):** Estimate the amount of GHGs associated with each effect. At this stage, calculation will be imprecise.
- **Trend (limited choice):** Evaluate whether emission associated with the effect could change over time. Consider structuring this as a set of limited choice options.
- **Trend-Rationale (narrative):** Explain why conditions might change, providing insights into the factors driving these changes.